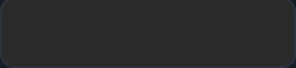


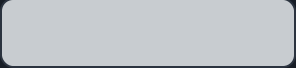
Scan colour + material → 3-D models (v0.80.0)

If the scans/FLIS state a colour, material or finish, the 3-D model now wears it — as a procedural metallic/matte/rubber finish.

Material → finish (parsed, verified)



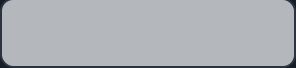
RUBBER · BLACK
matte



STEEL · ZINC
metallic



BRASS
warm gloss



STAINLESS
bright



OLIVE DRAB · PAINT
flat



RED · PLASTIC
semi-matte

Parse (material_feature)

- From the scan: COLOR · MATERIAL · SURFACE TREATMENT/FINISH.
- Military colours (olive drab/CARC/sand) + metals + polymers +
- platings. Plating/oxide overrides the colour.

FLIS-sourced, authoritative.

Render (gl3d material)

- Shader gains mat=[spec, shininess, metallic]; metals tint
- their own highlight, drop the plastic rim. Applied on open
- via /api/part_material; SVG fallback takes the colour.

Looks like the real material.

Honest scope

The scans give material WORDS, not surface photographs — so 'texture' here is a procedural finish (colour + metalness + roughness), not a photographic image map. That's what makes a steel bolt read metallic, a rubber hose matte-black, a brass fitting warm-glossy. Verified across 8 material/colour/finish cases; route in RUN-ALL-TESTS.bat. Additive, RPS-safe (default = original look; SVG fallback keeps the colour).